



HELIOS

How the System Works

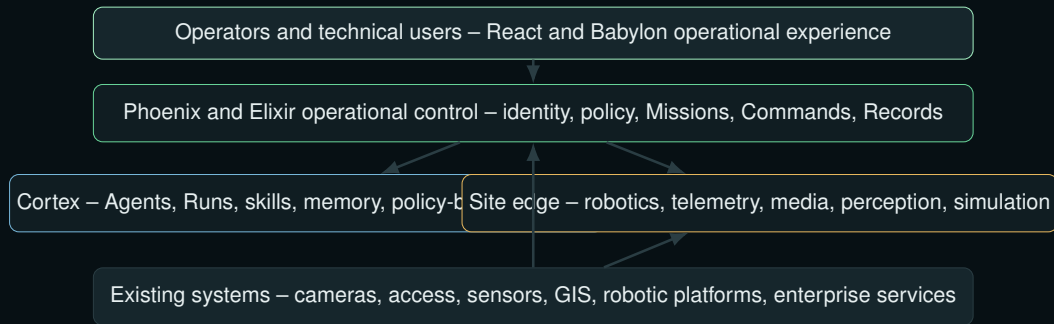
Operational world, Cortex, Missions, robotic assets, and resilient Site cells

CONTROLLED TECHNICAL OVERVIEW

Architecture and engineering evaluation edition

Version 0.1 – August 2026

System at a glance



Recommended baseline

Site-local operational continuity with central enterprise coordination. Existing systems remain authoritative for the facts and controls they own.

One authority path. Deliberate technical boundaries.

Operational control plane owns

Organization and Site identity; Operations; policy; Mission and Task lifecycle; resource reservation; Command admission; evidence relationships; audit; public interfaces.

Cortex owns

Agent identity and replaceable Runs; provider supervision; bounded tools; assessment; recommendations; memory retrieval; policy-permitted intent; structured requests.

Site and edge own

Vendor protocols; target execution; local rejection and failsafes; high-rate telemetry; media; perception; simulation; measured native workloads.

Paths that do not exist

Browser, scheduler, or Cortex directly operating a platform; simulation identity becoming physical execution; adapters creating Mission authority; client state becoming operational truth.

Constitution

Every effect-capable human, schedule, Cortex, policy, Agent, or tool request enters the same BEAM-owned admission path.

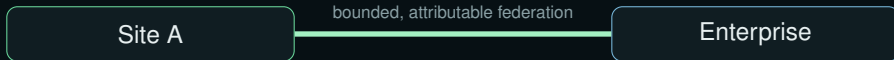
Resilient Site cells with enterprise coordination

SITE-LOCAL HELIOS

- ▶ Local operator access and durable state.
- ▶ Current operational world and approved Missions.
- ▶ Site Cortex and bounded local policy.
- ▶ Robotics, telemetry, media, and perception.
- ▶ Evidence buffering and local observability.
- ▶ Bounded operation during enterprise disconnection.

ENTERPRISE HELIOS

- ▶ Multi-Site posture and Attention.
- ▶ Organization policy and administration.
- ▶ Central Cortex and longer-horizon analysis.
- ▶ Cross-Site requests and resource coordination.
- ▶ Federated Records, memory, and learning.
- ▶ Approved package and policy distribution.



Disconnection never creates new authority. Reconnect reconciles existing facts and never duplicates effects.

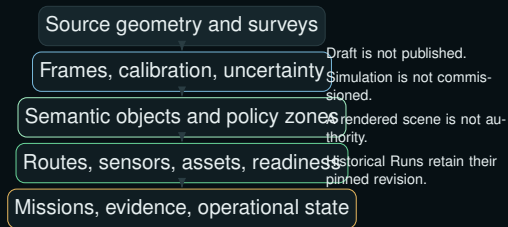
The operational world is authoritative context

World Revision

Source digests; geometry; coordinate frames; transforms; calibration; uncertainty; semantic objects; zones; routes; sensors; deployments; limitations; publication and commissioning state.

Presentation package

Babylon or another approved renderer consumes a derived package. Presentation may be optimized; authority cannot move into the renderer.



Mission and Command: intent to observed effect



Mission structure

Reusable Definition; immutable Revision; exact Proposal; target preparation; authorization snapshot; one admitted Run; independent Task Runs; attributable completion and Record.

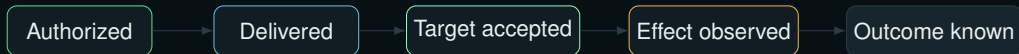
Command truth is orthogonal

Authority, delivery, target acceptance, observed effect, and Mission consequence remain separate. One dimension cannot silently advance another.

Unknown result

If a Command may have reached a target but the response is lost, Helios reconciles that attempt. It does not blindly issue another effect.

Why sent is not the same as happened



No blind replay

A lost response does not create permission to issue another physical effect.

One current controller

Operator, Cortex Run, Site process, and edge execution obey current generation or lease authority.

Record what was known

Intent, authority, delivery, acceptance, effect, uncertainty, intervention, and outcome remain attributable.

A missing receipt remains unknown. A stale controller is fenced. Reconnect reconciles the original attempt.

Cortex: hybrid intelligence with fenced Runs

Durable Agent

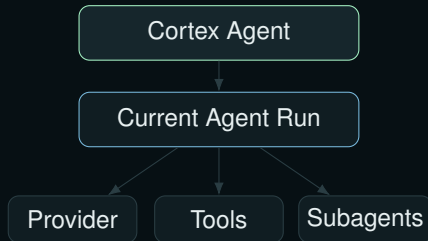
Accountable identity, role, scope, profile revision, authority ceiling, skill and tool policy, memory scope, and human supervisor.

Replaceable Agent Run

Generation, provider, approved context, skills, tools, budget, audience, consequence class, checkpoint, and lease. A newer generation fences an older Run.

Phase 3 autonomy

Current policy + evidence + exact Site and Mission class + readiness + eligible resources + preserved reserve coverage → ordinary Mission and Command admission → operator supervision.



Media, integration, identity, and trust

Media and perception

Camera or sensor → Site ingest → selected viewing and source-authoritative recording. Perception produces attributable detections, Tracks, observations, confidence, and provenance.

Cortex input

Structured semantic output, approved stills or clips, evidence references, source health, spatial context, uncertainty, and policy. Continuous raw video stays in the media path.

Integration patterns

Attributable events; snapshot plus delta; bounded source query; media reference; native command boundary; evidence export. Clients reload authoritative projections after gaps.

Identity and trust

Human, service, device, and Agent identities remain distinct. Service communication is mutually authenticated and encrypted. Browsers and model providers never receive reusable source or platform secrets.

Evidence rule

Reading, detection, Track, observation, association, Alert, Incident, evidence, assessment, recommendation, decision, Command, and outcome remain distinct attributable facts.

Degraded operation, delivery, and status

Failure behavior

- ▶ Source or media outage: unavailable with freshness; no negative inference.
- ▶ Cortex outage: native operator controls remain.
- ▶ Driver outage: resource degrades or becomes unavailable.
- ▶ Enterprise outage: bounded Site-local operation continues.
- ▶ Authority uncertainty: writes fail closed.
- ▶ Unknown effect: reconcile; never blind replay.

Visible delivery path

Foundation: identity, event, mode, and no-bypass controls.

Phase 1: operator-directed robotic Mission and Exercise.

Phase 2: Cortex assessment, recommendations, memory, learning.

Phase 3: policy-bounded autonomous observation.

Phase 4: multi-Agent and cross-Site coordination.

Current technical posture

CURRENT Foundations **IN DEVELOPMENT** Mission, Command, readiness, edge, operational Cortex **PLANNED**
Federation, autonomy, memory, learning, coordination

This overview withholds implementation-ready schemas, API inventories, database definitions, exact recovery logic, Site-specific network details, operational procedures, and proprietary source structure.